

Routeur: Cost-Aware Multi-Task LLM Routing Under a 10 ms CPU Budget

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July 2026

Abstract

Large language model routing is a constrained decision problem: a router should select the cheapest model that is sufficiently capable for each request, while adding negligible latency. We present Routeur, an open-source multi-task router that jointly predicts a five-level difficulty target, task, risk, and required capabilities. Two compact BERT students are dynamically quantized to INT8 and served with ONNX Runtime. The 4.4M-parameter economy profile reaches 49.15% exact and 92.68% adjacent validation accuracy, with 1.58% severe under-routing and a routed-cost ratio of 1.036 relative to the labelled oracle. The 11.3M-parameter accuracy profile reaches 51.23% exact and 93.38% adjacent accuracy, with 1.32% severe under-routing. On an Intel Xeon E5-1650 v3 server their p95 decision latencies are 1.61 and 3.62 ms. We also describe a critical limitation: under-routing and over-routing below 0.1% would require at least 99.8% exact accuracy, which cannot be supported by the subjective and boundary-noisy labels in the present corpus. Our evaluation therefore emphasizes measured cost-quality Pareto points and explicit failure rates rather than an unverifiable near-perfect claim.

1 Introduction

Routing every request to the most capable model wastes money and energy, while routing every request to a small model degrades quality. Prior work has framed this problem through preference prediction, benchmark-derived supervision, and learned model selection [1, 2]. The deployment constraint is often overlooked: if the router is slow or expensive, it consumes the benefit it is intended to create.

Routeur targets a strict p95 CPU decision latency below 10 ms. The system makes one semantic encoder pass, predicts four related targets, and applies a cost-aware model policy. It exposes two production profiles instead of one unqualified “best” model: an economy profile minimizes artifact size and cost, whereas an accuracy profile accepts a modest cost increase for better routing.

This work contributes: (i) a compact multi-task architecture; (ii) a reproducible data pipeline combining teacher labels, pairwise evidence, and public routing datasets; (iii) INT8 CPU deployment with measured latency on two older Xeon processors; (iv) explicit cost and asymmetric error metrics; and (v) an honest analysis of the statistical ceiling imposed by the current labels.

2 Problem Formulation

For prompt x , the router predicts a level $\hat{y} \in \{1, \dots, 5\}$ and then selects a callable model m from catalog $\mathcal{M}$. The labelled oracle level y is the cheapest tier believed to satisfy a quality threshold. Let $c(m)$ be normalized inference cost and $q(m, x)$ denote suitability. The deployment goal is approximately

$$\min_{m \in \mathcal{M}} c(m) \quad \text{subject to } q(m, x) \geq q_{\min}, \quad L_{95} < 10 \text{ ms.} \quad (1)$$

For a single five-way decision, exact, under-, and over-routing partition the sample space:

$$P(\hat{y} = y) + P(\hat{y} < y) + P(\hat{y} > y) = 1. \quad (2)$$

Consequently, requiring both directional errors below 0.1% implies exact accuracy above 99.8%. This identity is independent of model architecture. It is therefore invalid to report all three targets without an oracle whose agreement and stability support that precision.

We report exact accuracy, adjacent accuracy $P(|\hat{y} - y| \leq 1)$, severe under-routing $P(y - \hat{y} \geq 2)$, task accuracy, cost ratio to the labelled oracle, savings relative to always using level 5, and uncached single-request latency.

3 Data Construction

The final training mixture contains 49,984 normalized unique prompts. It combines four evidence types. First, teacher-labelled prompts provide difficulty, task, risk, and capability targets. Second, pairwise model outcomes reduce dependence on purely verbal difficulty judgments. Third, MMR Routing 20K contributes multi-turn conversations with independently observed weak-versus-strong model scores [5]. Fourth, targeted synthetic examples improve coverage of rare tasks and high-stakes language.

Exact normalized prompt overlap is removed across partitions. The 1,585-example teacher-gold split is used for checkpoint and profile selection, so we call it a validation set rather than a test set. A 3,000-example disjoint MMR set is kept as an external diagnostic. Its tier labels are derived from observed score gaps and do not define the same oracle as teacher-gold data. TwinRouterBench and CodeRouterBench are retained as additional distribution-shift diagnostics [6, 7].

This distinction matters. Difficulty levels compress a continuous, model- and policy-dependent trade-off into five bins. Adjacent tiers often represent nearly identical prompts, and teacher confidence is not a reliable proxy

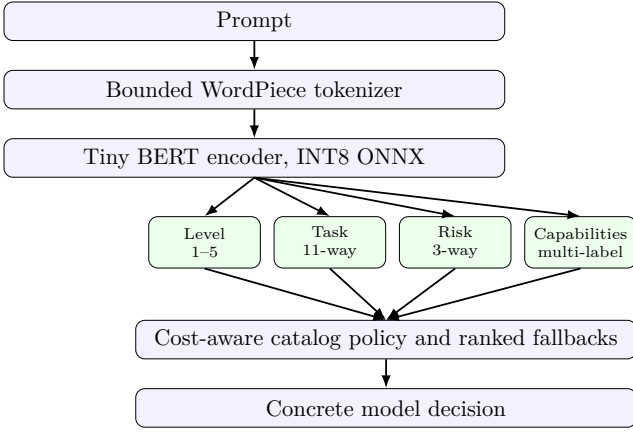


Figure 1: Router inference architecture. The learned heads feed a catalog policy; optional conservative guards can escalate high-stakes requests.

for label agreement. The corpus is useful for model selection but cannot certify a 99.8% exact router.

4 Architecture

Figure 1 summarizes inference. WordPiece tokenization is bounded to 128 tokens. A compact bidirectional Transformer produces a pooled representation h . Four heads predict level, task, risk, and multi-label capabilities:

$$p_y = \text{softmax}(W_y h + b_y), \quad (3)$$

$$p_t = \text{softmax}(W_t h + b_t), \quad (4)$$

$$p_r = \text{softmax}(W_r h + b_r), \quad (5)$$

$$p_c = \sigma(W_c h + b_c). \quad (6)$$

Training minimizes a weighted sum of cross-entropy losses and binary cross-entropy for capabilities. Shared representation learning is useful because task, risk, and capability requirements provide semantic supervision even when the precise level boundary is noisy.

The economy student uses the two-layer, 128-hidden, two-attention-head BERT Tiny configuration [4]; the accuracy student uses four layers, hidden size 256, and four attention heads. A six-layer MiniLM candidate [3] improved exact validation accuracy only marginally to 51.48%, but its 12.52 ms local p95 violated the deployment constraint and it was rejected.

5 Training and Compression

Training runs on an ephemeral H100 GPU using mixed precision, gradient clipping, best-checkpoint selection, and a persistent artifact volume. No GPU container is kept warm after the run. The two-layer model is trained for eight epochs; the four-layer model for six. The encoder and all heads are exported together to ONNX opset 17.

Dynamic signed INT8 quantization is applied to matrix multiplications. Dynamic quantization is a natural CPU choice for Transformer weights and activations whose ranges are computed at runtime [8]. Per-channel weights and reduced ranges are used for compatibility

Table 1: Validation cost-quality results. Lower is better for severe under-routing and cost ratio; higher is better otherwise.

Metric	Linear	Economy	Accuracy
Exact accuracy (%)	48.39	49.15	51.23
Adjacent accuracy (%)	91.92	92.68	93.38
Severe under-route (%)	2.78	1.58	1.32
Task accuracy (%)	59.50	58.30	61.83
Savings vs level 5 (%)	88.98	89.07	88.10
Cost/oracle	—	1.036	1.128
Artifact (MB)	52	5	12

with older AVX2 CPUs. ONNX Runtime is configured with one intra-op and one inter-op thread to reduce tail-latency variance. The resulting artifacts, including tokenizers, are approximately 5 MB and 12 MB.

An earlier hashed n-gram linear router is preserved as a reproducible baseline. An ensemble with the semantic model increased validation accuracy, but the gain did not persist on a held-out calibration partition; it was not promoted.

6 Cost-Aware Policy

The learned level is an internal compatibility signal rather than the final product decision. The policy ranks callable catalog entries using tier fit, task probabilities, required capability coverage, safety, public quality priors, relative price, and provider-diverse fallbacks. Let $s(m, x)$ be the weighted suitability score. A simplified view is

$$m^* = \arg \max_{m \in \mathcal{M}} [s(m, x) - \lambda_c \tilde{c}(m) - \lambda_o d(m, \hat{y})], \quad (7)$$

where $\tilde{c}$ is normalized price and d penalizes tier mismatch. Catalog shortlists avoid scanning every model for every request.

The default profiles rely on learned risk, safety, and capability heads. Two optional guard modes are available. A model-confirmed guard requires at least two learned high-risk signals before escalation. A lexical guard escalates any matched high-stakes phrase. The latter is intentionally conservative but caused substantial false-positive level-5 routing on validation data, so it is not the cost-optimized default.

7 Experimental Results

All latency measurements use 20 warm-ups and 1,000 unique prompts, which bypass the in-process LRU cache. Runs are sequential and use one ONNX Runtime thread. Table 1 reports the teacher-gold validation results. Exact values are available in each artifact’s `evaluation.json`.

The economy model is the strongest cost point: it is smaller than the linear baseline, improves exact and adjacent accuracy, nearly halves severe under-routing, and spends only 3.6% above the labelled oracle. The accuracy profile improves exact accuracy by another 2.08 percentage points and lowers severe under-routing, but spends 12.8% above the oracle. Thus neither profile dominates the other.

Table 2: Uncached single-prompt CPU p95 latency in milliseconds.

Machine	Economy	Accuracy	10 ms SLO
Xeon E3-1271 v3	2.84	4.76	pass
Xeon E5-1650 v3	1.61	3.62	pass

Both profiles satisfy the target with substantial margin (Table 2). The rejected six-layer candidate demonstrates why architecture selection must include measured tail latency rather than parameter count or average time alone.

On the disjoint MMR diagnostic, economy and accuracy exact scores are 33.00% and 37.47%, while severe under-routing is 39.27% and 34.33%. These poor absolute rates reveal a major oracle shift: MMR levels encode observed performance gaps between particular weak and strong models, whereas teacher-gold levels encode a general difficulty judgment. We report these results because they prevent a misleading in-distribution interpretation. They are evidence that future work must train and evaluate against the same operational quality threshold and model catalog.

8 Limitations and Threats to Validity

The principal limitation is label validity. Five discrete tiers are not natural ground truth, and a prompt’s cheapest sufficient model changes with the model catalog, price, decoding settings, and quality threshold. The validation set was used for checkpoint selection. It measures model-selection quality, not a final unbiased generalization estimate.

Second, cost figures use a normalized policy snapshot rather than audited bills from a randomized production trial. Prompt and completion length distributions can change the realized savings. Third, latency is measured on two x86 CPUs and does not cover all runtime versions, contention levels, or ARM deployments. Fourth, disabling a hard lexical guard reduces false-positive escalation and cost but makes safety depend on learned heads. High-stakes deployments should select a conservative mode and measure its quality-cost consequences on their own traffic.

The near-100% objective should be revisited only after constructing a stable oracle. A suitable protocol would run every candidate model on each prompt, score outputs with multiple independent judges and human adjudication near tier boundaries, repeat ambiguous trials, and reserve a never-touched temporal test set. Selective routing with abstention could then guarantee low directional error on the accepted subset, at the cost of escalating uncertain requests.

9 Reproducibility

The public repository contains data builders, training code, model cards, ONNX export and quantization, both production artifacts, a latency gate, evaluation metrics,

and continuous integration. Raw third-party datasets are downloaded from their publishers to preserve licenses and provenance. The command-line interface can select either profile and can override the safety guard mode. Tests validate calibration, routing, batching, policy behavior, and serialization.

10 Conclusion

Routeur demonstrates that semantic multi-task routing can fit comfortably inside a 10 ms CPU budget without a large runtime artifact. The economy profile offers the best measured cost point, while the four-layer profile offers the best in-distribution accuracy and severe-under-routing result. More importantly, the experiments expose the gap between optimizing a noisy five-tier teacher label and measuring true cheapest-sufficient-model decisions. Progress toward near-perfect routing now depends more on a stable, production-aligned oracle than on increasing encoder size.

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